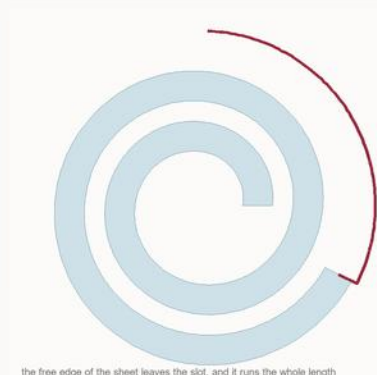
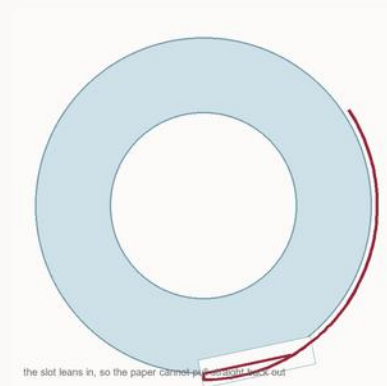


## Glass tip — two ways

Same job, opposite answers. One is a tube with a slot in it. The other is a rolled sheet, and the roll is the slot. Sections at right, with the paper in.



### Slotted tube

A slot the whole length, out both ends, raked seventy-eight degrees off the radius so it undercuts. Three millimetres of paper goes in. Thick wall, and a perforated screen across the bore.

|         |                                |
|---------|--------------------------------|
| Overall | <b>19 mm</b>                   |
| Outside | <b>ø 9 mm</b>                  |
| Bore    | <b>ø 5.0, 2.0 wall</b>         |
| Screen  | <b>seven ø 1.25 holes</b>      |
| Slot    | <b>0.75 wide, 3 mm of grip</b> |
| Mass    | <b>≈ 1.4 g</b>                 |

### Rolled sheet

No tube at all - a thin sheet rolled into a 9 mm cylinder, and the stock cut into lengths. The gap between the wraps is the slot, it runs the whole piece, and the free edge is where the paper goes in.

|         |                             |
|---------|-----------------------------|
| Overall | <b>19 mm</b>                |
| Outside | <b>ø 9 mm</b>               |
| Sheet   | <b>0.8 thick</b>            |
| Gap     | <b>0.55 between wraps</b>   |
| Wraps   | <b>about two and a half</b> |
| Mass    | <b>≈ 1.15 g</b>             |

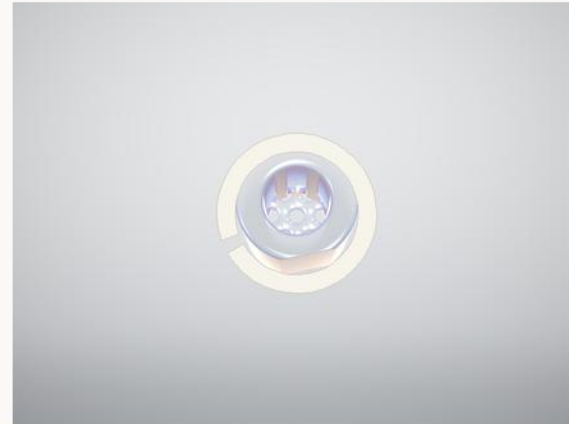
## Paper on, both ways

Three millimetres of the paper goes into the slot, along its whole length. Then it rolls.



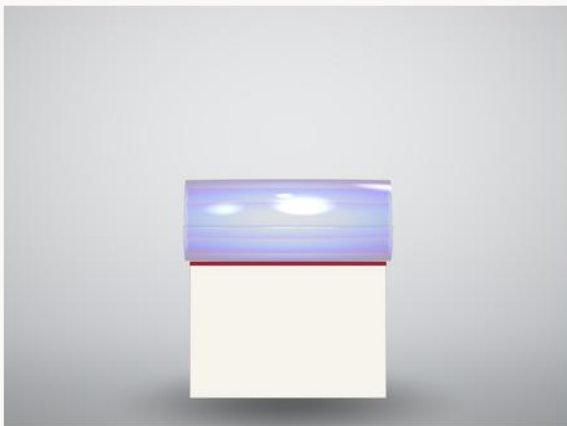
**Slotted tube**

paper in the slot



**Slotted tube**

wound on, end on



**Rolled sheet**

paper into the outer ring

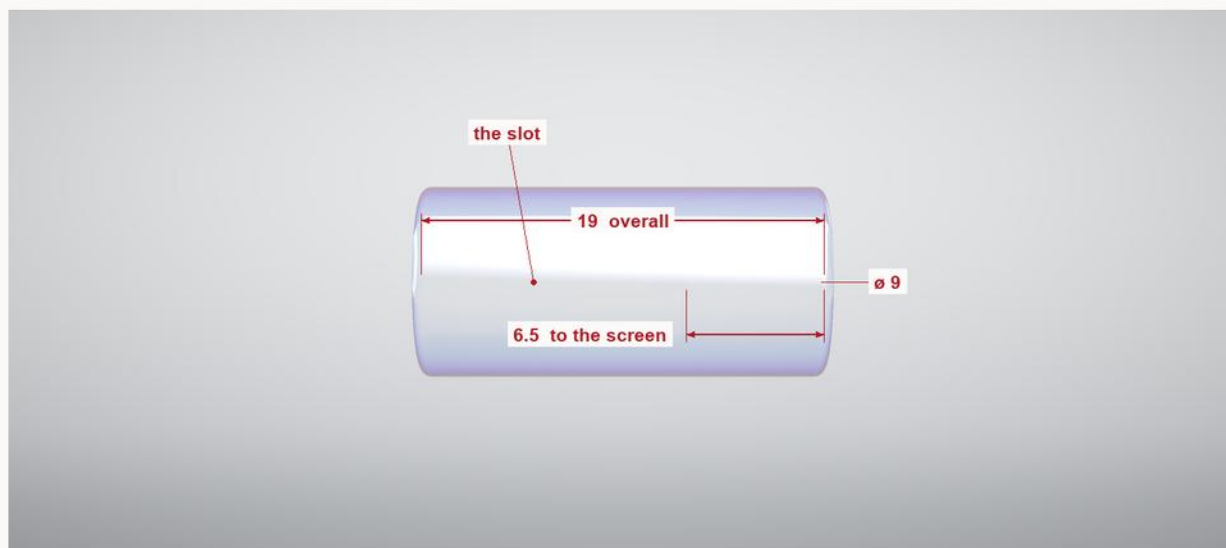


**Rolled sheet**

wound on, end on

## Glass tip

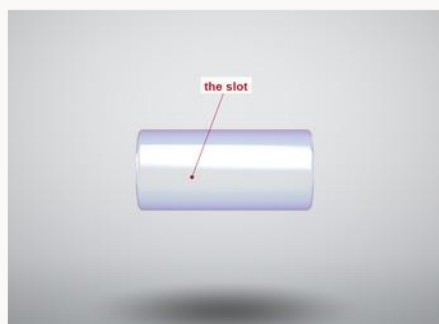
19 mm of clear 9 mm tube. Perforated screen across the bore, and a straight slot cut at a long oblique to start the roll.



Down the bore  
seven holes, one centre

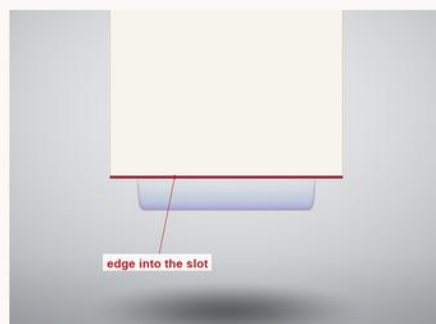
Overall length  
**19 mm**  
Outside diameter  
**9 mm**  
Bore / wall  
**6.4 / 1.3 mm**  
Screen  
**1.5 thick, 6.5 in**  
Screen holes  
**seven ø 1.25**  
Slot  
**0.75 wide, 3 mm of grip**  
Slot rake  
**68° off the axis**  
Glass  
**clear boro 3.3, ≈ 1.4 g**

### ROLLING



#### One

The slot runs the length of the tube, cut in on a lean.



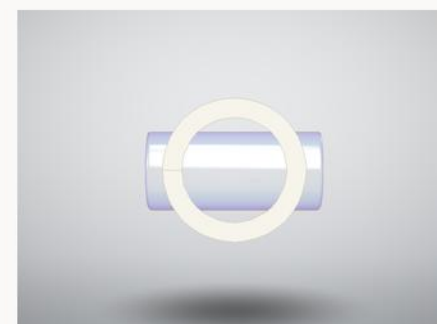
#### Two

Slip the leading edge of the paper into it. The slot holds the paper on its own.



#### Three

Roll the tip. The paper winds on, and the rake starts the wrap slightly off-square - the bias a cone wants.



#### Four

Wound on and filled. The screen keeps it from pulling through, and the tip washes and goes again.